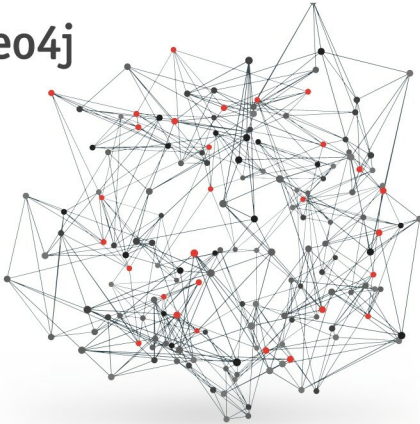


Finding Data in Graphs with Neo4j

Dubbed the database of the future, Neo4j is an open source graph database implemented in Java. This article is a teaser to whet your appetite for Neo4j and other graphical databases.



Technology is evolving so rapidly that very fast processors, high network speeds and agile computational algorithms are readily available and used, yet the performance of databases is of serious concern to those in the industry. Agility of business is important for the market, but what drags it down is that the schema for relational databases isn't designed for frequent changes. Moreover, data is evolving not only in volume and velocity, but also in complexity and interconnectedness. So RDBMSs have become a bottleneck as they were designed for tabular data, with a consistent structure and fixed schema. All these problems had led to some lateral thinking in search of a design which could be intuitive not just for developers, but for everyone familiar with the domain, and hence emerged Neo4j, the graphical data base.

Neo4j is an open source NoSQL graph database implemented in Java and Scala. It is sponsored by Neo Technology and widely used in production by eBay, Walmart, Telenor, etc. Some of the special features of Neo4j are:

- Implementation of the property graph model efficiently down to the storage level
- Materialising of relationships at creation time, resulting in

no penalties for complex runtime queries

- Constant time traversals for relationships in the graph both in depth and in breadth, due to efficient representation of nodes and relationships
- Written on top of JVM, it has compact storage and memory caching for graphs, resulting in efficient scale-up and billions of nodes in one database, even on moderate hardware

Matchmaking, network management, software analytics, scientific research, routing, organisational and project management, recommendations and social networks are some of the popular use cases of Neo4j. The Neo4j database can be downloaded from <http://neo4j.com/download> and for installation, follow the sequence: *Run -> Next -> Next and Finish*, just as with the VLC media player installation. This graph database comes with a Web browser based UI bound to <http://localhost:7474>. The simplest way of getting started is to use a Neo4j's database browser to execute your graph queries (written in Cypher) in a workbench-like fashion. Results are presented as either intuitive graph visualisation or as easy-to-read, exportable tables. The browser has some